

University of Calcutta
2ND Midterm M.Tech. Semester- 1 Examination, 2025

Paper Code & Name: CSC901 MATHEMATICAL FOUNDATION OF COMPUTER SCIENCE

Subject: Computer Science & Engineering

FULL MARKS 30

TIME 1 HR

Students can use T table and Z tables as per need.

1. Answer any four questions. (4x2.5=10)

- a. A fair coin is tossed repeatedly and independently. Find the expected number of tosses till the pattern HTH appears.
- b. Explain equilibrium in the Markov process with an example.

For a particular shop the daily demand of an item with associated probabilities is given below:

Daily demand	0	10	20	30	40	50
Probability	0.01	0.20	0.15	0.50	0.12	0.02

- c. For the sequence of random numbers (out of 100 random numbers generated between 00-99) used are as 25, 39, 65, 76, 12, 05, 73, 89, 19, 49 what will be the average daily demand?
- d. For the sequence of random numbers (out of 100 random numbers generated between 00-99) used are as 40, 19, 87, 83, 73, 84, 29, 09, 02, 20 what will be the average daily demand?
- e. Explain Continuous Parameter and Discrete State Space Stochastic Process with an example.

2. Answer any two questions. (2x10=20)

A. A Telecom service provider claims that individual customers pay on an average 400 rs. per month with standard deviation of 25 rs. A random sample of 50 customers bills during a given month is taken with a mean of 250 and standard deviation of 15. What to say with respect to the claim made by the service provider?

B. In the population, the average IQ is 100 with a standard deviation of 15. A team of scientists want to test a new medication to see if it has either a positive or negative effect on intelligence or not effect at all. A sample of 30 participants who have taken the medication has a mean of 140. Did the medication affect intelligence?

C. A vote is to be taken among the residents of a town and the surrounding county to determine whether a proposed chemical plant should be constructed. The construction site is within the town limits, and for this reason many voters in the county believe that the proposal will pass because of the large proportion of town voters who favor the construction. To determine if there is a significant difference in the proportions of town voters and county voters favoring the proposal, a poll is taken. If 120 of 200 town voters favor the proposal and 240 of 500 county residents favor it, would you agree that the proportion of town voters favoring the proposal is higher than the proportion of county voters? Use an $\alpha = 0.05$ level of significance.

D. Assume that a man's profession can be classified as professional, skilled labourer, or unskilled labourer. Assume that, of the sons of professional men, 80 percent are professional, 10 percent are skilled labourers, and 10 percent are unskilled labourers. In the case of sons of skilled labourers, 60 percent are skilled labourers, 20 percent are professional, and 20 percent are unskilled. Finally, in the case of unskilled labourers, 50 percent of the sons are unskilled labourers, and 25 percent each are in the other two categories. Assume that every man has at least one son, and form a Markov chain by following the profession of a randomly chosen son of a given family through several generations. Set up the matrix of transition probabilities. Find the probability that a randomly chosen grandson of an unskilled labourer is a professional man.

Advanced Algorithm

M. Tech, Semester I, Mid Semester Examination II,

Total marks: 30

Duration: 1 hour

1. Answer any 5 questions from the following:

[2 x 5= 10]

- h) Compute the prefix function for the string AABAACAABABC
- i) Compute the suffix function for the the string BBACBACBBAC
- j) Show the 1D range tree for the inputs 23, 27, 12, 45, 9, 56, 34, 87
- k) What is the advantage of using a skip list over binary search trees?
- l) Compare Trie Vs. Binary search tree data structure when searching a string of length m.
- m) For k dimensional orthogonal search query, name the most efficient data structure justifying its efficiency.
- n) What is the probability that a good pivot is chosen in the context of randomized Quicksort? Justify.

2. Answer any 2 questions from the following:

[10 x 2= 20]

- d) i) In a randomized closest pair algorithm, how and for what purposes randomization is used?

ii) Derive the expected case time complexity of insertion in a skip list. **[5+5]**
 - e) i) Derive the time complexity of divide and conquer closest pair algorithm.

ii) Derive the time complexity of KMP prefix function computation and KP matcher presenting the algorithms. **[4+6]**
 - f) i) For 2D orthogonal range searching using k-D tree, state and prove the time complexity of searching.

ii) Devise a randomized algorithm for binary search. Derive its expected case time complexity. **[6+4]**
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University of Calcutta
2nd Midterm M.Tech. Semester- I Examination, 2025

Subject: Computer Science & Engineering
Paper Code & Name: CSE901 & MACHINE LEARNING

Full Marks: 30

Time: 1:00 Hour

[2x5=10]

1. Answer any five of the following questions:

- i. What is batch normalization, and what are its advantages and disadvantages?
- ii. Derive the bias-variance trade-off for the squared error loss function. Explain what each term (bias, variance, irreducible error) represents intuitively and how they behave as model complexity increases.
- iii. What are hard margins and soft margins for SVMs? Give examples
- iv. Explain why PCA cannot identify independent components even if they exist in the dataset.
- v. If two features are perfectly correlated, what happens to the PCA components?
- vi. Compute the entropy of a node with class distribution (6 positive, 2 negative).
- vii. Give examples where decision trees and SVM will fail, and their justifications with alternative solutions.

2. i. How to select a Kernel for SVM? Explain the "kernel trick" in Support Vector Machines (SVMs). How does it allow learning a non-linear decision boundary without explicitly transforming the data into the high-dimensional feature space? Provide a simple example of a kernel function.
- ii. What are the terminating conditions while generating a decision tree? Give examples. Give some use cases where the decision tree generation process repeatedly uses a single feature.
- iii. What are the sustainability tests of the decision tree and SVM algorithms? Justify your answer.
- iv. Write down the basic steps of PCA with justifications. What are the limitations of PCA?

[5x4=20]

12:30 PM Sefsh
M. Tech

M.Tech 2nd Mid-Sem I Examination 2025
Subject: Wireless and Mobile Computing (CSE 903)

Full Marks: 30 Duration: 1 $\frac{1}{2}$ hour

1. Answer any five from the following: (5 x 2 = 10)

- a. "Link State routing may not be a suitable approach for a large network"- Comment.
- b. State the specific advantage of using MARCH protocol.
- c. State the exposed station problem.
- d. What is the function of SGSN and GGSN in GPRS network?
- e. Determine the data rate of TCH/H channel in GSM.
- f. What is the use of network slicing in 5G?
- g. How does eNB handles data and voice communication in 3G?

2. Answer any four from the following (4 x 5 = 20)

- a. State the purpose of Back-off algorithm used in MAC protocol. Compare the back-off strategies as used in MACA and MACA-W protocols.
 - b. State the hidden station problem and also illustrate the mechanism to address the problem by MACA. Comment on the performance of the said method.
 - c. Describe the DBTMA Protocol highlighting the advantage over MACA-W.
 - d. Discuss a frequency reuse algorithm. Which strategy can be used to handle sudden spike in demand for frequency in a particular cell.
 - e. Describe the logical channel structure in GSM, explaining the roles of traffic channels and control channels with suitable diagrams.
 - f. What do you mean by Cognitive Radio? What is the use of MIMO antenna in 5G? Discuss any 3 use cases suitable with 5G.
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